INCH-POUND

MIL-DTL-17/218A
05 October 2014
SUPERSEDING
MIL-C-17/218
29 September 1993

DETAIL SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE COAXIAL, 125 OHMS, M17/218-00001, UNARMORED, M17/218-00002, ARMORED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

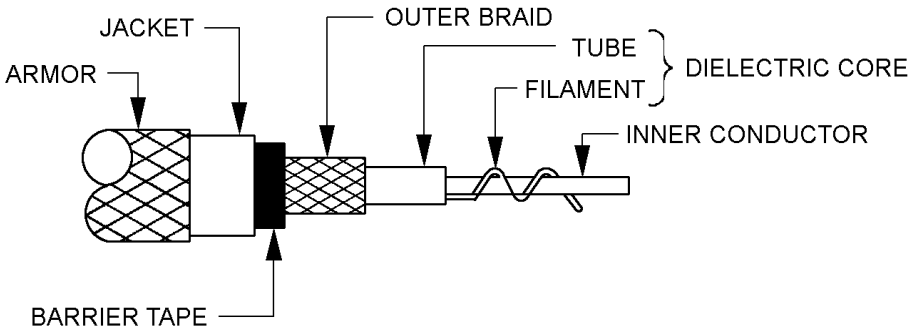

FIGURE 1. Configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Solid, copper-covered, steel wire. Diameter: .0253 inch ± .0010.
Dielectric core	Type A-3: Air-spaced polyethylene. A monofilament thread, .070 inch approx. diameter, with a lay of 1/2 inch approx.., under an extruded tube. Overall diameter: .285 inch ± .010. Alternate A continuous tube, .003 inch thick maximum under two continuous spiral fins, with a lay of 1-1/4 inches approx.., under an extruded tube of type A dielectric. Overall diameter: .285 inch ± .010.
Outer conductor	Single braid of AWG#33, bare, copper wire. Diameter: .340 inch maximum. Alternate Coverage: 94.9% nominal 94.9% nominal Carriers: 16 24 Ends: 12 8 Picks/inch: 4.3 ± 10% 6.5 ± 10%
Barrier type	A .001 inch thick polyester tape faced with a .002 inch thick layer of aluminum. The tape will be applied with a 50% lap, minimum. Aluminum face toward the outer conductor. Diameter: .350 inch, maximum.

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TABLE I. Description – Continued.

Jacket	Cross-linked polyolefin. Diameter: .405 inch ± .010.
Armor M17/218-00002 only	Single braid of aluminum-alloy wire. Diameter: .475 inch maximum.

ENGINEERING INFORMATION:

Continuous working voltage: 750 V rms, maximum.

Operating frequency: 1 GHz, maximum.

Velocity of propagation: 84 percent, nominal.

Power ratings: See figure 2.

Operating temperature range: -40° to +80°C, maximum.

Inner conductor properties:

DC resistance (maximum at +20°C): 4.4 ohms per 100 feet.

Elongation: 1 percent, minimum.

Tensile strength: 110 klbf/inch/inch, minimum.

The US Government preferred system of measurement is the metric SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound, the inch-pound units shall take precedence.

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination: Applicable.

Out-of-roundness: Not applicable.

Eccentricity: Not applicable.

Adhesion of conductors: Not applicable.

Aging stability: +98° ± 2°C.

Cold bend -40° ± 2°C.

Stress crack resistance: Not applicable.

Dimensional stability: Not applicable.

Contamination: Not applicable.

Flame propagation: Applicable.

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Acid gas generation: 2.0 percent, maximum.

Halogen content: 0.2 percent, maximum.

Immersion test:

Tensile strength, percent of unaged minimum: 50.

Elongation, percent of unaged minimum: 50.

Smoke index: 25 maximum.

Toxicity index: 5 maximum

Durometer hardness: (Type A) 80 minimum.

Weathering: Applicable.

Abrasion resistance: 75 cycles, minimum (jacket only).

Tear strength: 35 pounds per inch minimum.

Heat distortion: 30 percent maximum distortion.

Physical tests on unaged jacket:

Tensile strength: 1,300 psi. minimum.

Elongation: 160 percent minimum.

Physical tests on aged jacket:

Air oven:

Tensile strength, percent minimum: 60.

Elongation, percent minimum: 60.

Hot oil immersion:

Tensile strength, percent minimum: 50.

Elongation, percent minimum: 50.

Tensile strength and elongation: 1,300 psi, 160 percent minimum.

Weight: 8.8 pounds per 100 feet maximum (M17/218-00001). 13.8 pounds per 100 feet maximum (M17/218-00002).

Electrical:

Spark test: 5,000 V rms, minimum.

Voltage withstanding: 3,000 V rms, minimum.

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Insulation resistance: Not applicable.

Corora extinction voltage: Not applicable.

Characteristic impedance: 125 ohms ± 6.

Attenuation: 5.5 dB per 100 feet, maximum at .4 GHz.

Structural return loss: Not applicable.

Capacitance: 11.0 pF per foot, maximum.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise: Not applicable.

Time delay: Not applicable.

Part or Identifying Number (PIN): M17/218-00001 and -00002.

Referenced documents. This document references MIL-DTL-17.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA-CC

Review activities:
Army – AR, AT, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19, 99

(Project 6145-2014-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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